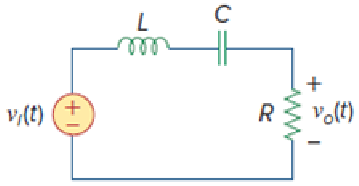
Chapter 14, Problem 11PP

Problem

Design a band-pass filter of the form in Fig. 14.35 with a lower cutoff frequency of 20.1 kHz and an upper cutoff frequency of 20.3 kHz. Take R = 30 kΩ. Calculate L, C, and Q.

Figure 14.35: A band-pass filter.



Step-by-step solution

Step 1 of 5

Refer to circuit diagram in Figure 14.35 in the text book.

The lower cutoff frequency is $f_1 = 20.1\text{ kHz}$, and upper cutoff frequency is $f_2 = 20.3\text{ kHz}$.

Calculate the Band width in rad/s.

$$\begin{aligned} B &= \omega_2 - \omega_1 \\ &= 2\pi f_2 - 2\pi f_1 \end{aligned}$$

Substitute 20.3 kHz for f_2 and 20.1 kHz for f_1.

$$\begin{aligned} B &= 2\pi\left(20.3\times10^3\right) - 2\pi\left(20.1\times10^3\right) \\ &= 2\pi\left(0.2\times10^3\right) \\ &= 1256.64\text{ rad/s} \end{aligned}$$

Step 2 of 5

Write the expression for band width in terms of resistor and inductor.

$$B = \frac{R}{L}$$
$$L = \frac{R}{B}$$

Substitute 1256.64 rad/s for B and $30\text{ k}\Omega$ for R.

$$\begin{aligned} L &= \frac{30\times10^3}{1256.64} \\ &= 23.87\text{ H} \end{aligned}$$

Therefore, the value of inductor, L is $\boxed{23.87\text{ H}}$.

Step 3 of 5

Calculate the resonant frequency.

$$\omega_0 = 2\pi\sqrt{f_2 f_1}$$

Substitute 20.3 kHz for f_2 and 20.1 kHz for f_1.

$$\begin{aligned}\omega_0 &= 2\pi\sqrt{(20.3\times10^3)(20.1\times10^3)}\\ &= 2\pi\sqrt{408.03\times10^6}\\ &= 2\pi(20.19\times10^3)\\ &= 126.92\text{ krad/s}\end{aligned}$$

Step 4 of 5

Write the expression for resonant frequency in terms of inductor and capacitor.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$LC = \frac{1}{\omega_0^2}$$

$$C = \frac{1}{\omega_0^2 L}$$

Substitute 126.92 krad/s for ω_0 and 23.87 H for L,

$$\begin{aligned}C &= \frac{1}{(126.92\times10^3)^2(23.87)}\\ &= \frac{1}{384.5\times10^9}\\ &= 2.6\text{ pF}\end{aligned}$$

Therefore, the value of capacitor, C is $\boxed{2.6\text{ pF}}$.

Step 5 of 5

Write the expression for quality factor.

$$Q = \frac{1}{\omega_0 RC}$$

Substitute 126.92 krad/s for ω_0, $30\text{ k}\Omega$ for R and 2.6 pF for C.

$$\begin{aligned}Q &= \frac{1}{(126.92\times10^3)(30\times10^3)(2.6\times10^{-12})}\\ &= \frac{1}{9.899\times10^{-3}}\\ &= 101\end{aligned}$$

Therefore, the value of quality factor is $\boxed{101}$.